

Lean PS: Typed Correspondence, Witness Persistence, and a Verified Kernel for Principia Symbolica Book V

Working paper scaffold for the AGI-26 correspondence program

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Abstract

We describe a machine-checked architecture connecting three formally distinct domains: Kripke–Joyal forcing under refinement, covariance of physical force laws, and selected arithmetic and conditional-system claims from *Principia Symbolica* Book V. The connection is a shared typed commutation interface, not an identification of semantics. The current receipt contains 1737 verified declarations across the analysis stack. Of these, 233 declarations map to 104 Book V anchors. The Book V atlas contains 75 theorem-like claims, of which 0 remain explicitly unmapped. We present the proof boundary, negative controls, and a receipt-driven pedagogy pipeline that refuses to promote paper prose to kernel fact.

1 The correspondence is an interface

The common object is a commuting condition. In the total setting, equivariance states

$$f(g \cdot x) = g \cdot f(x).$$

In the relational forcing setting, an admissible refinement step is preserved by the forcing predicate. The two constructions instantiate one interface but differ in totality, equality strength, and semantics. The formal architecture therefore retains domain, process schema, observer scope, adjudication mode, and transport-loss class as independent coordinates.

This separation blocks two invalid inferences. First, agreement across observers does not imply persistence under refinement. Second, commutation of a physical square does not reflect membership in the Lorentz group without nondegeneracy. Both failures have machine-checked finite or algebraic countermodels.

2 Witness preservation and M-Pers

A witness system assigns certificate types $W(o, p, a)$ to observer o , condition p , and atom a , together with restriction

$$q \preceq p \implies W(o, p, a) \rightarrow W(o, q, a).$$

Stability is inhabitation of the certificate type. Hence local persistence is derived:

$$\text{Stable}(o, p, a) \wedge q \preceq p \implies \text{Stable}(o, q, a).$$

Cross-observer transport is a second structure with a named naturality law. Material persistence may use both axes, but horizontal agreement alone does not establish the vertical theorem.

3 Book V proof families

3.1 Balanced memory and reciprocity

Balanced two-step memory uses

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Lean checks the eigenpair $(\varphi, 1)$, uniqueness of the positive root of $x^2 = x + 1$, and convergence of consecutive Fibonacci ratios. Weighted reciprocity uses

$$T_w = \begin{pmatrix} 1 & w \\ 1 & 0 \end{pmatrix}, \quad \lambda_w = \frac{1 + \sqrt{1 + 4w}}{2}.$$

The kernel proves $\lambda_w^2 = \lambda_w + w$, the corresponding eigenvector, $\lambda_1 = \varphi$, sub-golden rates for $0 < w < 1$, and super-golden rates for $w > 1$. Claims about ethics, decency, or agency remain interpretations of this conditional model.

3.2 Fracture and product spectra

For a nonzero equal-coordinate plane diagonal, Lean proves

$$\frac{\|v\|_1}{\|v\|_2} = \sqrt{2}, \quad \frac{\|v\|_1}{\|v\|_\infty} = 2.$$

Separate countermodels show that fixing the memory regime does not determine the norm regime and fixing the norm regime does not determine memory. Thus the symbolic-geometric interface is represented by typed product coordinates rather than a single sovereign classification. The finite-support kernel now includes positive finite p , the infinity regime, sharp equality cases, effective support dimension, and finite symbolic-path decoherence.

3.3 Conditional thermodynamics

Free energy and viability are definitions. Conservation, entropy production, fixed-point equivalence, persistence, and covenant stability are represented by structures carrying their laws. Lean proves consequences from those fields; it does not install Book V's modeling laws as global axioms. For example, nonnegative entropy rate follows from the explicit comparison

$$\rho_{\text{reflection}} \leq \sigma_{\text{production}},$$

and persistence follows only after supplying a positive-energy persistence bridge.

Coherence balance is likewise an energy-budget identity, not zero energy. When drift and reflection rates agree, Lean proves only that the energy rate equals the external rate. Conservation requires the additional closed-budget hypothesis; a concrete compiled countermodel refutes conservation from balance alone.

4 Receipt-driven pedagogy

The build emits a theorem receipt containing declaration name, source module, axiom dependencies, and a **no-sorry** gate. A second projection maps Principia anchors to exact, conditional, partial, or definitional coverage. Matt consumes these claims; Ellie composes a semantic lesson; the transcript

gate rejects ungrounded numbers, status words, and guarded universalizations. The same semantic program is suitable input for textbook, slides, audio, or video renderers. Rendering is downstream of proof and does not strengthen it.

5 Open program and workshop ingestion

The immediate proof queue is:

1. deployed Surface witness types and a construction of M-Pers for actual transition rules;
2. trajectory-response hypotheses for Book V adaptation, kept separate from life classification and metabolic regulation;
3. spectral hypotheses sufficient for the MAD–MAP–MAS classification;
4. compact or measurable path-space extensions when later claims require more than the finite symbolic-decoherence kernel;
5. functorial witness restriction and observer transport.

A future AGI-26 workshop paper can be ingested by mapping each statement to this receipt discipline: definition, proved theorem, conditional theorem, measurable conjecture, modeling postulate, or open debt. Shared diagrams do not merge their domains. The weakest adequate composition remains the governing rule.

6 Observer lowering and complementary half-bridges

The quantum correspondence is directional. Its two Gleason-facing halves are

normalized source $\longrightarrow$ Hermitian density $\longrightarrow$ resolved readout,
 certified readout laws $\longrightarrow$ Hermitian or sesquilinear representation.

The lower line represents observer-visible values; it does not invert the upper line. Global phase supplies a kernel-certified collision at the seam, and the frame and resolution countermodels block unconditional reverse reconstruction. Finite partial trace is separately certified as a reduction preserving trace and represented local expectations.

The temporal arrow is independently formal. The Cacophony-facing kernels certify finite-support norm fracture and a strict diagonal path-cost gap; ordered stage costs telescope; and a minimizing JKO step pays its transport displacement from free-energy decrease. Summable stage bounds in a complete observer metric, or positive Lyapunov descent paying each increment, certify convergence. The interpretive step is narrower and functional: mapping physical quantum decoherence and noise onto this general directed transport-and-loss geometry. Like a weather map, the correspondence is operational and testable without asserting a kernel identity between the mapped domains.

7 Reproducibility boundary

The paper compiles only after `book5_lean_coverage.py` generates the count macros above. The JSON coverage packet lists every mapped anchor and every unmapped claim-bearing Book V node.

The Lean root prints the axiom closure of each headline declaration. Numerical Lorentz and Plomp–Levelt witnesses remain separate conformance artifacts. Poems and interpretive prose are preserved as authored content and are not represented as proof obligations.